

PPR
Technical Catalogue

berkeplastik



Factories with high technology

Berke Plastik represents Turkey with investments on PVC-U, PP-R, PE-X, metal, heating product groups and high technological product facilities. Berke Plastik is in the list of the first 1000 exporters in Turkey. Besides, it is an international mark with its experience and quality.

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Worldwide Recognition



Berke Plastik is known worldwide

Berke Plastik has been ranked among the best marks in Turkey and the world with its fund of knowledge that is based on its product quality, production technology, customer focused service and long-established experience.

• PP-R; Polypropylene Random Copolymer Type - 3

This material is a proven, high performance random copolymer that enables the production of top quality solutions for hot & cold water piping applications. Thanks to its high exceptional heat stability, extraction resistance, stable processing characteristics, and ease of welding and installation, random copolymer enables pipes with reliable performance to be manufactured

- Extremely long life at least 50 years
- Taste and odour neutral
- Excellent weldability
- Good chemical resistance
- Bacteriologically neutral
- Physiologically harmless

• Physical, Thermal and Mechanical Properties

Properties	Testing methods	Unit	Values
Physical properties			
Density at 23 °C	ISO 1183	g /cm3	0.9
Melt flow index (MFI) 190 °C / 5 kg	ISO 1133	g/10 min	<0.8
Melt flow index (MFI) 230 °C / 2.16 kg	ISO 1133	g/10 min	<0.5
Linear expansion coefficient	ISO 8078	K-1	1.5x10-4
Heat conductivity	ISO 8078	WK-1m-1	0.23
Surface resistance	ISO 8078	Ω	>1012
Elasticity module	ISO 8078	N/mm2	800
Thermal properties			
Melting point	DSC	°C	146-150
Subjective heat	Calorimeter	Kj/kgK	1.73
Coeficient of linear expansion	ASTM D 696	mm/m(°C)	0.15
Deflection temperature under load - 1.8 Mpa	ISO 75A-1, -2	°C	46
VICAT softening point	ISO 306	°C	132
Mechanical properties			
Tensile stress at yield, at 50 mm/min	ISO 527-1, -2	Mpa	25
Tensile stress at yield, at 50 mm/min		%	%
Tensile stress at yield, at 50 mm/min		13	>500
Charpy impact strenght (0°C)	ISO 179	15J	Fara rupere

• Long service life

With all water carrying pipes, resistance to internal pressure is an important factor affecting long life characteristics.

To assure optimal life performance, straight and curved pipes have been subjected to extensive hydrostatic pressure testing, in the laboratory, at a variety of temperatures.

• Long-lasting performance

Random copolymer that is used for production of Berke pipes and fittings can withstand temperatures up to 70° C without losing shape, and handle short duration temperatures of up to 100° C.

Combined with good chemical resistance and impact strength, Berke pipes can simply be installed and forgotten about for at least 50 years!

Particularly important for domestic water installations, is the fact that random copolymer type 3 is physiologically harmless, and taste and colour neutral.

• Excellent stability

In use, the formulation of Berke pipes is such that it offers a high molecular weight, plus excellent mechanical properties. Additional benefits include high heat stability as well as excellent resistance to extraction.

Berke pipes and fittigs conform to DIN 8077/78, EN ISO 15874 -1,2,3 and DIN 16962 standarts.

• Easy installation, cost-effective in use

Whatever the complexity of a domestic water installation, Berke pipes have the ability to be shaped to conform to even most complex layouts.

Berke pipes easy weldability also makes them faster and simpler to install.

• Measuring hydrostatic pressure performance

Hydrostatic pressure is calculated according to the below formula :

$$P = \sigma \times \frac{2e_{min}}{(d_{em} - e_{min})}$$

P	= internal pressure ,MPa
d_{em}	= outside diameter of the pipe, mm
e_{min}	= minimum wall thickness of the pipe, mm
σ	= hydrostatic stress, MPa
1 Mpa	= 10 bar

- **Service life DIN 8077 (SF = 1.5 PP-R)**

Temperature °C	Services Life	Pipe Series (S) - Standart Dimension Rate (SDR)							
		5	11	3.2	7.4	2.5	6	2	5
		PN 10		PN 16		PN20		PN25	
		Pressure (bar)							
20	1	15.0		23.7		29.9		37.7	
	5	14.1		22.3		28.1		35.4	
	10	13.7		21.7		27.4		34.5	
	25	13.2		21.0		26.4		33.3	
	50	12.9		20.4		25.7		32.4	
40	1	10.8		17.1		21.6		27.2	
	5	10.1		16.0		20.2		25.4	
	10	9.8		15.5		19.6		24.7	
	25	9.4		15.0		18.8		23.7	
	50	9.2		14.5		18.3		23.1	
60	1	7.7		12.2		15.4		19.4	
	5	7.1		11.3		14.3		18.0	
	10	6.9		11.0		13.9		17.5	
	25	6.6		10.5		13.3		16.7	
	50	6.4		10.2		12.9		16.2	
70	1	6.5		10.3		12.9		16.3	
	5	6.0		9.5		12.0		15.1	
	10	5.8		9.2		11.6		14.6	
	25	5.0		8.0		10.0		12.7	
	50	4.2		6.7		8.5		10.7	
80	1	5.4		8.6		10.8		13.7	
	5	4.8		7.6		9.6		12.1	
	10	4.0		6.4		8.1		10.2	
	25	3.2		5.1		6.5		8.1	
95	1	3.8		6.1		7.6		9.6	
	5	2.6		4.1		5.2		6.5	

• Standarts and guidelines

DIN 8077	Polypropylene (PP) pipes. PP-H. PP-B. PP-R. PP-RCT dimensions.
DIN 8078	Polypropylene (PP) pipes. PP-H. PP-B. PP-R. PP-RCT general quality requirements and testing
DIN 16962	Pipe fittings and joint assemblies for polypropylene (PP) pressure pipes. Part 5- General Quality Requirements and Testing.
DIN 1988	Part 6- Injection Moulded Elbows for socket-welding Dimension. Part 9 Injection Moulded reducers and nippels for socket welding Dimension.
ISO 3213	Drinking water supply systems- Part 1 General, Part 2 Materials, components, appliances, design and installation.
ISO 10508	Polypropylene (PP) pipes effect of time and temperature on the expected strength.
EN ISO 15874	Plastics piping systems for hot and cold water installations -- Guidance dor classification and design
DVGW 544	Plastic piping systems for hot and cold water installations polypropylene (PP) - Part 1 - general, Part 2 -pipe, Part 3 - fittings, Part 5 - fitness for purpose of the system, Part 7 -guidance for the assessment of conformity
DVS 2207	Plastic pipes in the drinking water installation; Requirements and testing

• Metal Component Compatibility

As with all polypropylenes prolonged exposure to copper can cause damage to the properties of random copolymer Type-3

Where metal inserts are to be used in an installation the recommended choice is nickel or chromium-plated brass components in order not to harm raw material properties. All the metal inserts that are used in production of Berke metal fittings are nickel or chromium-plated brass components that does not harm the raw material.

• Chemical Resistance

As with all PP pipes it is advisable that substances such as oils, waxes and bitumen should be kept away from the pipes.

Chemical or Product	Concentration Up to 40%	Temperature °C		
		20	60	100
Acetic acid		S	S	-
Acetic acid	50%	S	-	-
Acetic acid, qlacial	>96%	S		NS
Acetic anhydride	100%	S	L	-
Acetone	100%	S	-	-
Aceptophenone	100%	S	S	-
Acrylonitrile	100%	S	L	-
Air		S	-	S
Allyl alcohol	100%	S	S	-
Almond oil		S	S	-
Alum	Sol	S	-	-
Ammonium acetate	Sat.Sol	S	S	-
Ammonium fluoride	Up to 20%	S	S	-
Ammonium hydrogen carbonate	Sat. Sol	S	S	-
Ammonium metaphoSpheate	Sat.Sol	S	S	S
Ammonium nitrate	Sat.Sol	S	S	S
Ammonium perSulphate	Sat.Sol	S	S	-
Ammonium Sulphide	Sat.Sol	S	S	-
Amly acetate	100%	L	H	H
Amly alcohol	100%	S	S	S
Aniline	100%	S	S	-
Apple juice		S	-	-
Barium bromide	Sat.Sol	S	S	S
Barium carbonate	Sat.Sol	S	S	S
Barium hydroxide	Sat.Sol	S	S	S
Barium Sulphide	Sat.Sol	S	S	S
Benzoic acid	Sat.Sol	S	S	-
Benzyl alcohol	100%	S	L	-
Boron trifluoride	Sat.Sol	S	-	-
Boromine. qaS		NS	NS	NS
Boromine. liquid	100%	NS	NS	NS

Chemical or Product	Concentration Up to 40%	Temperature °C		
		20	60	100
Butanol	100%	S	L	L
Butyl acetate	100%	L	NS	NS
Butyl qlycol	100%	S	-	-
Butyl phenolS	Sat.Sol	S	-	-
Butyl phthalate	100%	S	L	L
Calcium carbonate	Sat.Sol	S	S	S
Calcium chlorate	Sat.Sol	S	S	-
Calcium chloride	Sat.Sol	S	S	S
Calcium hydroxide	Sat.Sol	S	S	S
Calcium hypochlorite	Sol	S	-	-
Calcium nitrate	Sat.Sol	S	S	-
Carbon dioxide, dry qaS		S	S	-
Carbon dioxide, wet qaS		S	S	-
Carbon diSulphide	100%	S	NS	NS
Carbon monoxSide. qaS		S	S	-
Carbon tetrachloride	100%	NS	NS	NS
CaStor oil	100%	S	S	-
CauStic Soda	Up to 50%	S	L	L
Chlorine, dry qaS	100%	NS	NS	NS
Chlorine, liquid	100%	NS	NS	NS
Chloroacetic acid	Sol	S	-	-
Chloroethanol	100%	S	-	-
Chloroform	100%	L	NS	NS
ChloroSulphonic acid	100%	NS	NS	NS
Chrom alum	Sol	S	S	-
Chromic acid	Up to 40%	S	L	NS
Citric acid	Sat.Sol	S	S	S
Coconut oil		S	-	-
Copper (II) chloride	Sat.Sol	S	S	-
Copper (II) nitrate	Sat.Sol	S	S	S
Copper (II)	Sat.Sol	S	S	-
Corn oil		S	L	-
CottonSeed oil		S	S	-
Cyclohexane	100%	S		-
Cyclohexanol	100%	S	L	-
Cyclohexaone	100%	L	NS	NS

Chemical or Product	Concentration Up to 40%	Temperature °C		
		20	60	100
Dextrin	Sol	S	S	-
DextroSe	Sol	S	S	S
Dibutyl phtalate	100%	S	L	NS
Dichloroacetic acid	100%	L	-	-
Dichloroethylene (A and B)	100%	L		-
Diethanolamine	100%	S	-	-
Diethyl ether	100%	S	L	-
Diethylene qlycol	100%	S	S	-
Diqlycolic acid	Sat.Sol	S		-
DiiSooctyl	100%	S	L	-
Dimethyl amine, qaS		S	S	-
Dimethyl formamide	100%	S	S	-
Dioctyl phthalate	100%	L	L	
DiStilled water	100%	S	S	S
Ethanolamine	100%	S	S	S
Ethyl acetate	100%	L	NS	NS
Ferric chloride	Sat.Sol	S	S	S
Formic acid	10%	S	S	L
Formic acid	85%	S	NS	NS
FructoSe	Sol	NS	NS	NS
Fruit juice		S	S	S
GaSoline, petrol (aliphatic hydrocarbonS)		NS	NS	NS
Gelatine		S	S	-
GlucoSe	20%	S	S	S
Glycerine	100%	S	S	S
Glycolic acid	30%	S	-	-
Hexane	100%	S	L	-
Hydrochloric acid	Up to 20%	S	S	S
Hydrochloric acid	30%	S	L	L
Hydrochloric acid	From35 to 36%	S	-	-
Hydrofluoric acid	Dil.Sol	S	-	-

Chemical or Product	Concentration Up to 40%	Temperature °C		
		20	60	100
Hydrofluoric acid	40%	S	-	-
Hydroqen	100%	S	-	-
Hydroqen chloride, dry qaS	100%	S	S	-
Hydroqen peroxide	Up to 30%	S	L	-
Hydroqen Sulphide, dry qaS	100%	S	S	0
Iodine in alcohol		S	-	-
ISopropvl alcohol	100%	S	S	S
ISopropvl ether	100%	L	-	-
Lactic acid	Up to 90%	S	H	
Lanoline		S	L	-
LinSeed oil		S	S	S
MaqneSium carbonate	Sat.Sol	S	S	S
MaqneSium chloride	Sat.Sol	S	S	S
MaqneSium hydroxide	Sat.Sol	S	S	
MaqneSium Sulphate	Sat.Sol	S	S	-
Maleic acid	Sat.Sol	S	S	-
Mercury (II) chloride	Sat.Sol	S	S	-
Mercury (II) cyanide	Sat.Sol	S	S	-
Mercury (1) nitrade	Sol	S	S	-
Mercury	100%	S	S	-
Methyl acetate	100%	S	S	-
Methyl amine	Up tp 32%	S	-	-
Methyl bromide	100%	NS	NS	NS
Methyl ethyl ketone	100%	S	-	-
Methylene cloride	100%	L	NS	NS
Milk		S	S	S
Monochloroacetic acid	>85%	S	S	-
Nickel chloride	Sat.Sol	S	S	-
Nickel nitrate	Sat.Sol	S	S	-
Nickel Sulphate	Sat.Sol	S	S	-
Nitric acid	From 40 to 50%	L	NS	NS
Nitric acid, fujminq (with nitroqen dioxide)		NS	NS	NS

Chemical or Product	Concentration Up to 40%	Temperature °C		
		20	60	100
Oleic acid	100%	S	L	-
Oleum (Sulphuric acid with 60% of S03)		S	L	-
Olive oil		S	S	L
Oxalic acid	Sat.Sol	S	L	NS
Oxvqen.qaS		S	-	-
Paraffin oil (FL65)		S	L	NS
Peanut oil		S	S	-
Peppermint oil		S	-	-
Perchloric acid	(2N) 20%	S	-	-
Petroleum ether		L	L	-
Phenol	5%	S	S	-
Phenol	90%	S	-	-
PhoSphine. qaS		S	S	-
PhoSphoruS oxychloride	100%	L	-	-
Picric acid	Sat.Sol	S		-
PotaSSium bicarbonate	Sat.Sol	S	S	S
PotaSSium borate	Sat.Sol	S	S	-
PotaSSium bromated	Uo to 10%	S	S	-
PotaSSium bromide	Sat.Sol	S	S	
PotaSSium carbonate	Sat.Sol	S	S	
PotaSSium chlarate	Sat.Sol	S	S	
PotaSSium chloerite	Sat.Sol	S	S	
PotaSSium chromate	Sat.Sol	S	S	
PotaSSium cyanide	Sol	S	-	
PotaSSium chloerite	Sat.Sol	S	S	
PotaSSium dichromate	Sat.Sol	S	S	S
PotaSSium ferricyanide	Sat.Sol	S	S	-
PotaSSium fluride	Sat.Sol	S	S	-
PotaSSium iodide	Sat.Sol	S	-	-
PotaSSium nitrate	Sat.Sol	S	S	-
PotaSSium pechlorite	10%	S	S	-
PotaSSium permanqanate	(2N) 39%	S	-	-
PotaSSium perSulphate	Sat.Sol	S	S	-
Propionic acid	>50%	S	-	-
Pyridine	100%	L	-	-

Chemical or Product	Concentration Up to 40%	Temperature °C		
		20	60	100
Sea water		S	S	S
Silver nitrate	Sat.Sol	S	S	L
Sodium acetate	Sat.Sol	S	S	S
Sodium benzoate	35%	S	L	
Sodium bicarbonate	Sat.Sol	S	S	S
Sodium carbonate	Uo to 50%	S	S	L
Sodium chlorite	20%	S	L	NS
Sodium dichromate	Sat.Sol	S	S	S
Sodium hydroqen carbonate	Sat.Sol	S	S	S
Sodium hydroqen Sulphate	Sat.Sol	S	S	-
Sodium hvdroqen Sulphite	Sat.Sol	S	-	-
Sodium hypochlorite	5%	S	S	-
Sodium hypochlorite	10%-15%	S	-	-
Sodium hypochlorite	20%	S	L	-
Sodium metaphoSpate	Sol	S	-	-
Sodium nitrate	Sat.Sol	S	S	-
Sodium perorate	Sat.Sol	S	S	-
Sodium phiSohate (neutral)		S	S	S
Sodium Silicate	Sol	S	S	-
Sodium Sulphate	Sat.Sol	S	S	-
Sodium Sulphide	Sat.Sol	S	-	-
Sodium Sulphite	40%	S	S	S
Sodium thioSulphate (hypo)	Sat.Sol	S	-	-
Say-bean oil		S	L	
Succinic acid	Sat.Sol	S	S	-
Sulphur acid	From 10 to 30%	S	S	-
Sulphuric acid	50%	S	L	L
Sulphuric acid	96%	S	L	NS
SulphurouS acid	Up to 30%	S	H	-
Tartaric acid	Sat.Sol	S	S	-
Tetralin	100%	NS	NS	NS
Thiophene	100%	S	L	-
Tin (IV) chloride	Sol	S	S	-
Tin (II) chloride	Sat.Sol	S	S	-
Toulene	100%	L	NS	NS

Chemical or Product	Concentration Up to 40%	Temperature °C		
		20	60	100
Trichloroacetic acid	Up to 50 %	S	S	-
Trichloroethylene	100%	NS	NS	NS
Triethanolamine	Sol	S	-	-
Turpentine		NS	NS	NS
Urea	Sat.Sol	S	S	-
Vinegar		S	S	S
Water brackiSh, mineral, portable		S	S	S
WineS		S	S	-
Xylene	100%	NS	NS	NS
YeaSt	Sol	S	S	S
Zinc Sulphate	Sat.Sol	S	S	-

S : Satisfactory
L : Limited
NS : Not Satisfactory
Sat. sol : Saturated aqueous solution prepared at 20°C
Sol : Aqueous solution at a concentration higher than 10% but not saturated
Dil. sol : Dilute aqueous solution at a concentration equal to or lower than %10
Work. sol : Aqueous solution having the usual concentration for industrial use

• Thermal expansion and support intervals of Berke pipes

Pipes elongate when heated. These changes in longitude should especially be regarded during installation of hot water installations.

To prevent the unwanted bendings the pipes should be supported at designated intervals. Freely selected fixed support should be arranged so as to balance the length changes via changing the direction of the flow in the installation.

If it is possible to balance such an expansion as in the case of straight pipe lines in between the two fixed supports resilience balancers are applied. During the assembly of the resilience balancers longitudinal changes caused by the functioning of the installation and by the changes in the environmental temperatures should be taken into account.

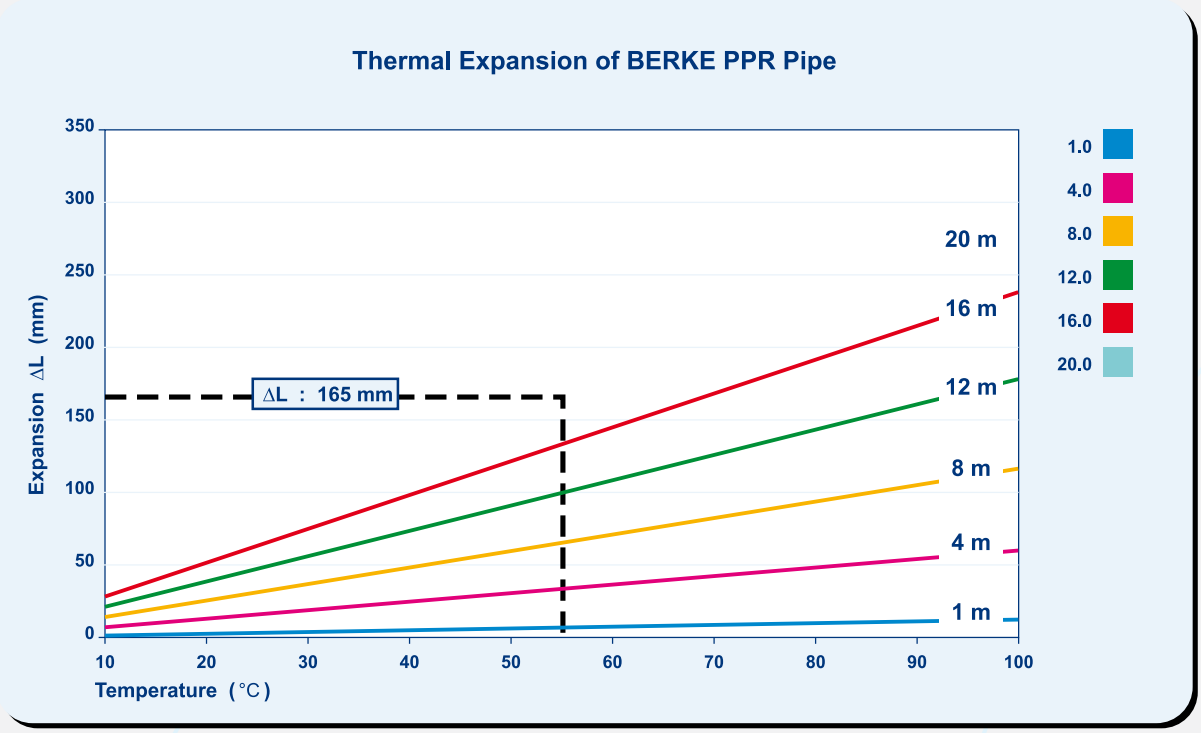
• Thermal expansion of Berke PP-R pipe

Calculation of thermal expansion is as follows:

$$\Delta L = L * \Delta T * \lambda$$

where

ΔT : variation of working temperature in Kelvin degrees (K) or Celsius (°C)
ΔL : variation of length in mm
L : initial length of the pipe in m
λ : coefficient of linear thermal expansion. The value of λ is $1.5 * 10^{-4}$ (K^{-1}) for pp-r pipe.



Pipe length (m)	Temperature variation AT in K									
	10	20	30	40	50	60	70	80	90	100
	Linear Expansion AL (mm)									
1.0	1.5	3.0	4.5	6.0	7.5	9.0	10.5	12.0	13.5	15.0
4.0	6.0	12.0	18.0	24.0	30.0	36.0	42.0	48.0	54.5	60.0
8.0	12.0	24.0	36.0	48.0	60.0	72.0	84.0	96.0	108.0	120.0
12.0	18.0	36.0	54.0	72.0	90.0	108.0	126.0	144.0	162.0	180.0
16.0	24.0	48.0	72.0	96.0	120.0	144.0	168.0	192.0	216.0	240.0
20.0	30.0	60.0	90.0	120.0	150.0	180.0	210.0	240.0	270.0	300.0

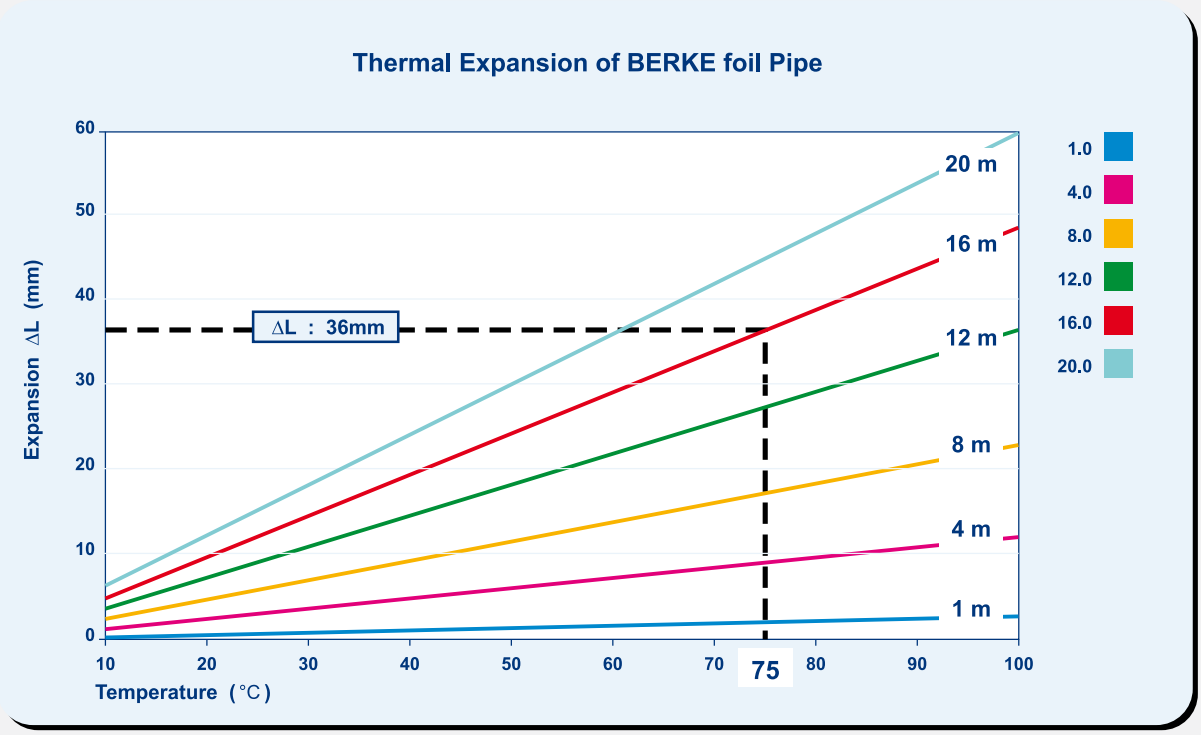
• Thermal expansion of Berke PP-R foil pipe

Calculation of thermal expansion is as follows:

$$\Delta L = L * \Delta T * \lambda$$

where

ΔT : variation of working temperature in Kelvin degrees (K) or Celsius (°C)
ΔL : variation of length in mm
L : initial length of the pipe in m
λ : coefficient of linear thermal expansion. The value of λ is $0.3 * 10^{-4}$ (K^{-1}) for pp-r foil tubes.



Pipe length (m)	Temperature variation ΔT in K									
	10	20	30	40	50	60	70	80	90	100
	Linear Expansion ΔL (mm)									
1.0	0,30	0,60	0,90	1,20	1,50	1,80	2,10	2,40	2,70	3,00
4.0	1,20	2,40	3,60	4,80	6,00	7,20	8,40	9,60	10,80	12,00
8.0	2,40	4,80	7,20	9,60	12,00	14,40	16,80	19,20	21,60	24,00
12.0	3,60	7,20	10,80	14,40	18,00	21,60	25,20	28,80	32,40	36,00
16.0	4,80	9,60	14,40	19,20	24,00	28,80	33,60	38,40	43,20	48,00
20.0	6,00	12,00	18,00	24,00	30,00	36,00	42,00	48,00	54,00	60,00

• Thermal expansion of Berke PP-R fiberglass pipe

Calculation of thermal expansion is as follows:

$$\Delta L = L * \Delta T * \lambda$$

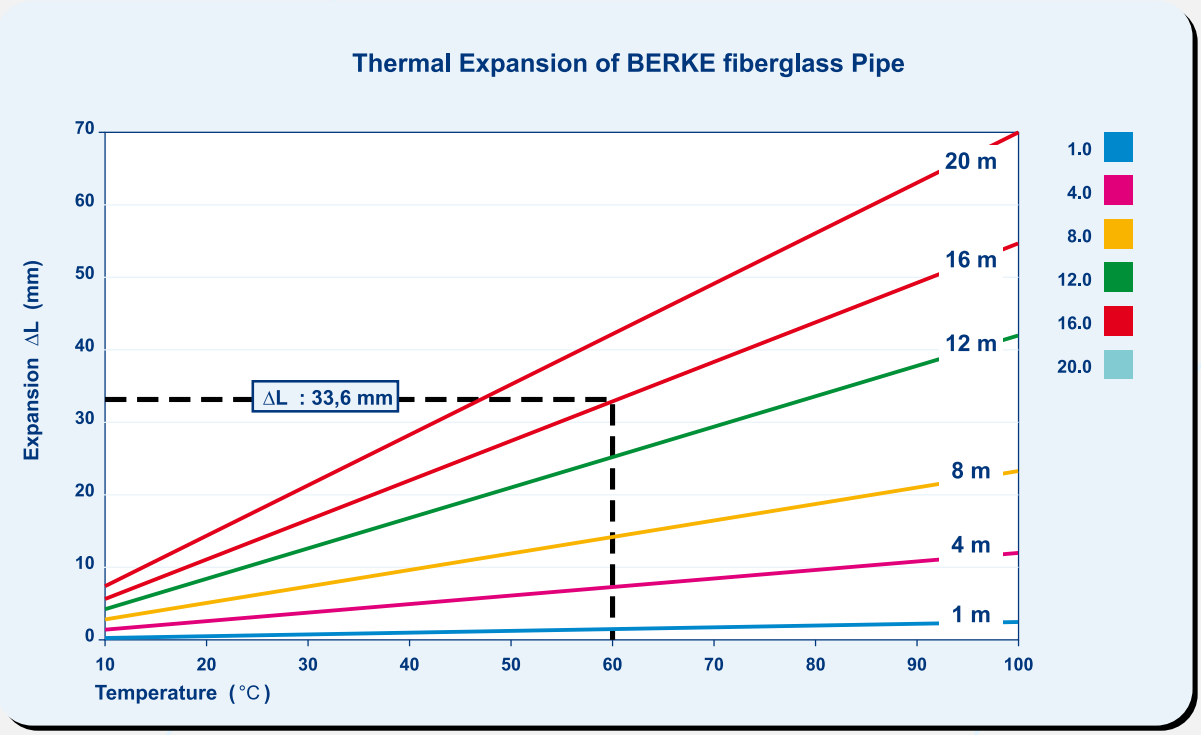
where

ΔT : variation of working temperature in Kelvin degrees (K) or Celsius (°C)

ΔL : variation of length in mm

L : initial length of the pipe in m

λ : coefficient of linear thermal expansion. The value of λ is 0.35 * 10^{-4} (K^{-1}) for pp-r fiberglass pipe.



Pipe length (m)	Temperature variation ΔT in K									
	10	20	30	40	50	60	70	80	90	100
	Linear Expansion ΔL (mm)									
1.0	0,35	0,70	1,05	1,40	1,75	2,10	2,45	2,80	3,15	3,50
4.0	1,40	2,80	4,20	5,60	7,00	8,40	9,80	11,20	12,60	14,00
8.0	2,80	5,60	8,40	11,20	14,00	16,80	19,60	22,40	25,20	28,00
12.0	4,20	8,40	12,60	16,80	21,00	25,20	29,40	33,60	37,80	42,00
16.0	5,60	11,20	16,80	22,40	28,00	33,60	39,20	44,80	50,40	56,00
20.0	7,00	14,00	21,00	28,00	35,00	42,00	49,00	56,00	63,00	70,00

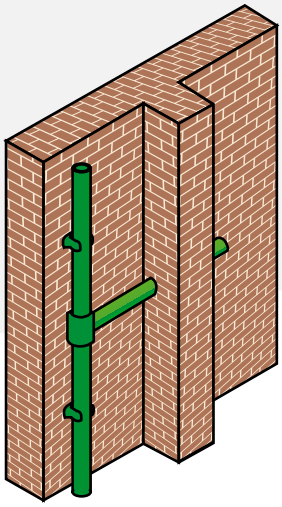
• Support intervals

Berke PP-R pipe SDR:6 – SDR:7.4 (PN20 – PN16)

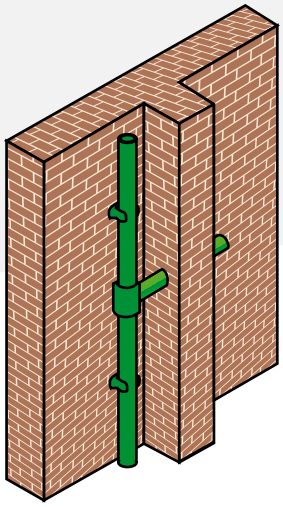
Temperatur ΔT (K)	Pipe diameter d (mm)								
	20	25	32	40	50	63	75	90	110
	Support intervals in cm								
20	60	70	90	100	120	140	150	160	180
30	60	70	90	100	120	140	150	160	180
40	60	70	80	90	110	130	140	150	170
50	60	70	80	90	110	130	140	150	170
60	50	60	70	80	100	110	120	140	160
70	50	60	70	80	90	100	110	120	140

• Type of Assembly for Installation of the flat

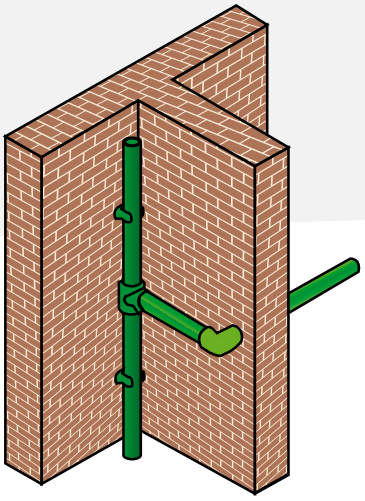
To allow for the subsequent linear expansion of the pipes regarding the connections at the entrances of flats, one of the techniques below should be used for making the connections.



A: This connection can be made at particular a distance from the wall.

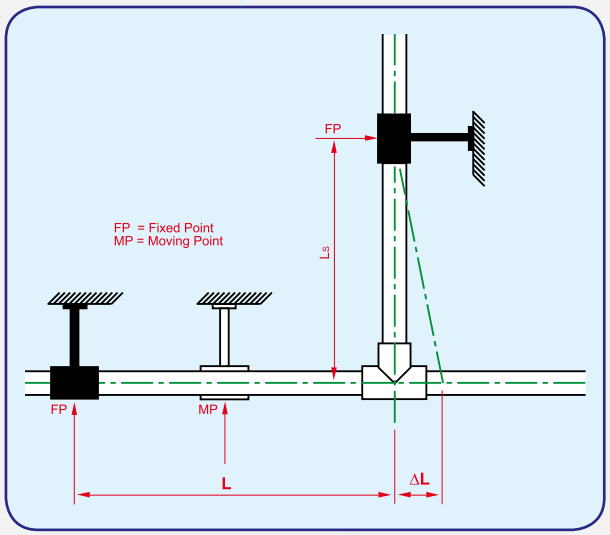


B: The connection can be made via allowing awide hole at the point where the pipe enters the flat.



C: The pipe can be inserted into the entrance point of the flat, via an “L” formed arm.

• Minimum distance for connection allowing for the expansion during joint connection



$$Ls = K\sqrt{d \times \Delta L}$$

Example :

L = 5m	ΔL = L*ΔT*λ	Ls = K √d x ΔL
ΔT = 50	ΔL = 5*50*0,15	Ls = 15√40*37,5
Ød = 40mm	ΔL = 37,5 mm	Ls = 580 mm
Ls = ?		

Ls : length of the flexible pipe side, mm
d : outer diameter of the Berke pipe, mm
ΔL : variation of length, mm
K : 15 (material based constant of Berke pipe)

• Length of the flexible pipe side is calculated with the following Formula:

Thermal expansion is calculated with the below stated Formula:

$$Ls = K\sqrt{d \times \Delta L}$$

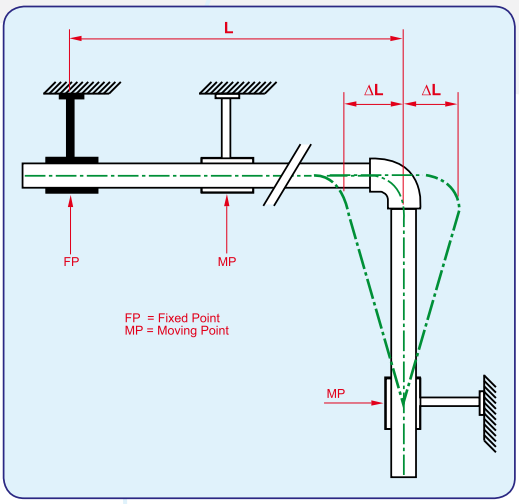
Ls : length of the flexible pipe side, mm
d : outer diameter of the Berke pipe, mm
ΔL : variation of length, mm
K : 15 (material based constant of Berke pipe)

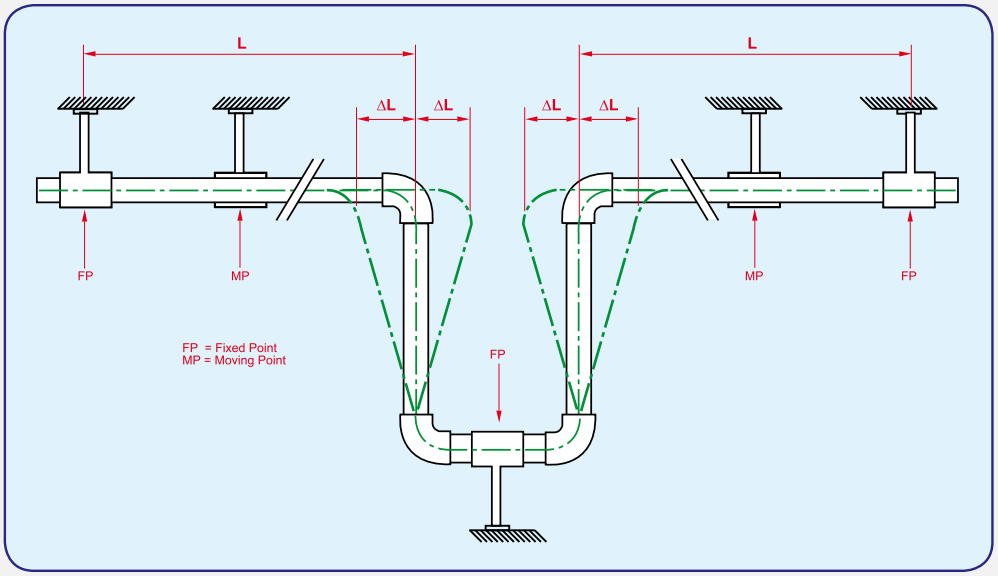
• Calculation of thermal expansion

Thermal expansion is calculated with the below stated Formula:

$$\Delta L = \lambda \times \Delta T \times L$$

Ls : variation of working temperature in Kelvin (K) or Celcius (°C)
d : variation of length in mm
ΔL : initial length of the pipe in m
K : coefficient of linear thermal expansion.
The value of λ is 1,5 * $10^{-4}(K^{-1})$ for pp-r pipe.





Example :

1- Calculation of thermal expansion

Temperature difference between cold water and enviroment

Input			Requirement		
λ	=	0.15 mm/m - K	ΔL	=	λ x ΔT x L
L	=	2.0 m	ΔL	=	0.15 x 40 x 2.0 = 12mm
ΔT	=	40K (°C)			

2- Calculation of the shortest flexible length

Input			Requirement		
d	=	40 mm	Ls	=	K * √ d * ΔL
ΔL	=	12 mm	Ls	=	15 * √40 * 12 = 328

• Insulation

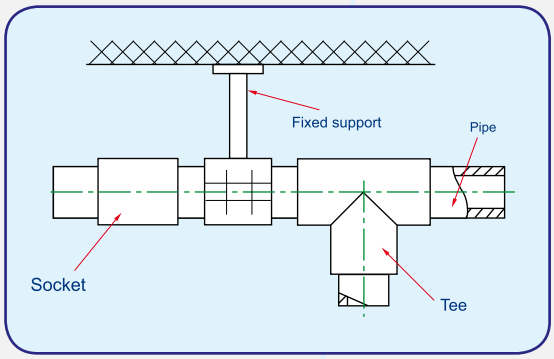
Pipe Outer Diameter	Insulation Thickness in BERKE Pipes	
	λ = 0,035 W/mK	λ = 0,04 W/mK
20 x 3.4 mm	16.5 mm	21.0 mm
25x4.2 mm	16.8 mm	20.2 mm
32 x 5.4 mm	16.9 mm	32.2 mm
40x6.7 mm	24.5 mm	31.0 mm
50x8.3 mm	25.0 mm	30.2 mm
63 x 10.5 mm	35.0 mm	42.0 mm
75 x 12.5 mm	41.0 mm	51.0 mm
90 x 15.0 mm	48.5 mm	60.5 mm
110x 18.3 mm	60.0 mm	74.2 mm

• Assembly Principles

Radial and axial expansion of the pipe assembled outdoors should not be prevented. To obtain good results fixed supports should be chosen in such way that they do not damage the outer surface of the pipe.

• Fixed Support

Fixed supports are used to fix the pipe at a certain points against undesired pipe movements. Fixed supports should be stronger compared to the sliding supports. Fittings are used to construct fixed supports. At the points where direction changes fixed supports should not be used. The distance between the fixed supports should be chosen in such a way that pipe thermal expansion is not affected.



For insulated BERKE pp-r pipes installed outdoor, thermal expansion should be calculated within the planperiod.

• Welding principles of Berke pipes and fittings

Preparation of the welding machine

1- Welding machine is heated up to 260°C. When the control light goes off at 260°C welding process can be started. Pipe should be cut to the appropriated length perpendicular to the pipe axis. Welding distance should be marked from the pipe.

If its is an al-foiled pipe

2- A special peeler is used to take off the co-extruded PP layer and Al-foil

3- Welding surfaces of pipes and fittings should be clean. If neccessary welding surfaces should be cleaned with alcohol and dried with a dry cloth. During welding neither the pipe nor the fittings should be moved.

Welding

4- Pipes and fittings should be heated. When the heating process ends they should be taken out quickly and joined by pressing one to another axially without twisting. Welding machine should be cleaned after every operation for the next use.

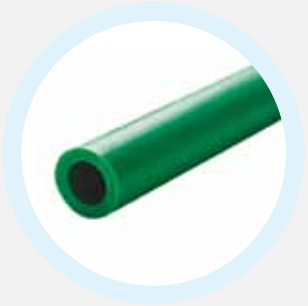


PP-R Product List

PP-R PIPE PN 10

ماسورة حراري ضغط ١٠ بار

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4010.10.020	20	100 m
1.4010.10.025	25	80 m
1.4010.10.032	32	40 m
1.4010.10.040	40	32 m
1.4010.10.050	50	20 m
1.4010.10.063	63	16 m
1.4010.10.075	75	12 m
1.4010.10.090	90	8 m
1.4010.10.110	110	4 m



PP-R PIPE PN 16

ماسورة حراري ضغط ١٦ بار

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4010.16.020	20	100 m
1.4010.16.025	25	80 m
1.4010.16.032	32	40 m
1.4010.16.040	40	32 m
1.4010.16.050	50	20 m
1.4010.16.063	63	16 m
1.4010.16.075	75	12 m
1.4010.16.090	90	8 m
1.4010.16.110	110	4 m



PP-R PIPE PN 20

ماسورة حراري ضغط ٢٠ بار

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4010.20.020	20	100 m
1.4010.20.025	25	80 m
1.4010.20.032	32	40 m
1.4010.20.040	40	32 m
1.4010.20.050	50	20 m
1.4010.20.063	63	16 m
1.4010.20.075	75	12 m
1.4010.20.090	90	8 m
1.4010.20.110	110	4 m



PP-R SUPER PIPE PN 25 (MIDDLE-FOILED/FOIL WITH HOLES/TRANSPARENT INLAYER)

ماسورة مع طبقة المنيوم في الوسط ٢٥ بار



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4024.25.020	20	100 m
1.4024.25.025	25	80 m
1.4024.25.032	32	40 m
1.4024.25.040	40	32 m
1.4024.25.050	50	20 m
1.4024.25.063	63	16 m

PP-R SUPEROXY PIPE PN 25 (MIDDLE-FOILED/ FOIL WITHOUT HOLES/GREEN INLAYER)

ماسورة مع طبقة المنيوم في الوسط ٢٥ بار ملونة داخلي خضرى



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4026.25.020	20	100 m
1.4026.25.025	25	80 m
1.4026.25.032	32	40 m
1.4026.25.040	40	32 m
1.4026.25.050	50	20 m
1.4026.25.063	63	16 m
1.4026.25.075	75	12 m
1.4026.25.090	90	8 m
1.4026.25.110	110	4 m

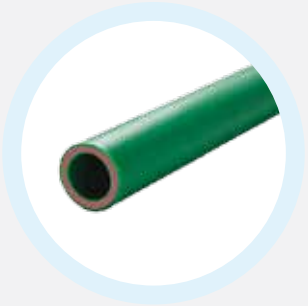
PP-R UV RESIST PIPE PN 25

ماسورة حراري مقاوم لأشعة الشمس ٢٥ بار



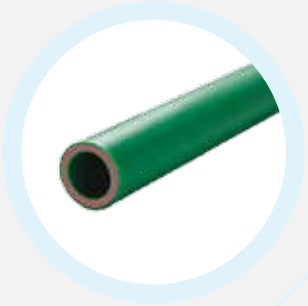
الرمز Code	المقاس ملم d (mm)	التعبئة Package Standart
1.4022.25.020	20	100 m
1.4022.25.025	25	80 m
1.4022.25.032	32	40 m
1.4022.25.040	40	32 m
1.4022.25.050	50	20 m
1.4022.25.063	63	16 m
1.4022.25.075	75	12 m
1.4022.25.090	90	8 m
1.4022.25.110	110	4 m

PP-R GF PIPE PN 20 ماسورة حراري متعدد الطبقات فيبر زجازي ٢٠ بار



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4030.20.020	20x2,8	100 m
1.4030.20.025	25x3,5	80 m
1.4030.20.032	32x4,4	40 m
1.4030.20.040	40x5,5	32 m
1.4030.20.050	50x6,9	20 m
1.4030.20.063	63x8,6	16 m
1.4030.20.075	75X10,3	12 m
1.4030.20.090	90X12,3	8 m
1.4030.20.110	110X15,1	4 m

PP-R GF PIPE PN 25 ماسورة حراري متعدد الطبقات فيبر زجاجي ٢٥ بار



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4030.25.020	20x3,4	100 m
1.4030.25.025	25x4,2	80 m
1.4030.25.032	32x5,4	40 m
1.4030.25.040	40x6,7	32 m
1.4030.25.050	50x8,3	20 m
1.4030.25.063	63x10,5	16 m
1.4030.25.075	75x12,5	12 m
1.4030.25.090	90x15	8 m
1.4030.25.110	110x18,3	4 m

PP-R STRAIGHT CONNECTOR جلبة حراري



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.12.020	20	600
1.4040.12.025	25	350
1.4040.12.032	32	210
1.4040.12.040	40	100
1.4040.12.050	50	60
1.4040.12.063	63	36
1.4040.12.075	75	24
1.4040.12.090	90	16
1.4040.12.110	110	8

PP-R ELBOW 90°

كوع زاوية حراري



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.13.020	20	500
1.4040.13.025	25	300
1.4040.13.032	32	140
1.4040.13.040	40	70
1.4040.13.050	50	35
1.4040.13.063	63	16
1.4040.13.075	75	12
1.4040.13.090	90	4
1.4040.13.110	110	3

PP-R ELBOW 45°

كوع فاتح حراري



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.14.020	20	500
1.4040.14.025	25	350
1.4040.14.032	32	150
1.4040.14.040	40	70
1.4040.14.050	50	50
1.4040.14.063	63	20
1.4040.14.075	75	12
1.4040.14.090	90	10
1.4040.14.110	110	4

PP-R TAIL ELBOW 45°

كوع فاتح ذكر حراري



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.16.020	20	300
1.4040.16.025	25	200

PP-R TAIL ELBOW 90°

كوع زاوية ذكر حراري



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.15.020	20	300
1.4040.15.025	25	200

PP-R TE

قسام حراري



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.18.020	20	300
1.4040.18.025	25	175
1.4040.18.032	32	80
1.4040.18.040	40	50
1.4040.18.050	50	25
1.4040.18.063	63	14
1.4040.18.075	75	10
1.4040.18.090	90	4
1.4040.18.110	110	2

PP-R UNEQUAL TE

قسام نقاصة حراري



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.19.012	20x25x20	200
1.4040.19.014	25x20x20	175
1.4040.19.016	25x20x25	175
1.4040.19.018	25x25x20	175
1.4040.19.020	32x20x20	100
1.4040.19.022	32x20x25	100
1.4040.19.024	32x20x32	80
1.4040.19.026	32x25x20	100
1.4040.19.028	32x25x25	100
1.4040.19.030	32x25x32	100
1.4040.19.042	40x20x40	60
1.4040.19.050	40x25x40	60
1.4040.19.058	40x32x40	50
1.4040.19.074	50x20x50	35
1.4040.19.084	50x25x50	35
1.4040.19.094	50x32x50	30
1.4040.19.104	50x40x50	30
1.4040.19.116	63x20x63	20
1.4040.19.128	63x25x63	20
1.4040.19.140	63x32x63	20
1.4040.19.152	63x40x63	16
1.4040.19.154	63x50x63	16
1.4040.19.168	75x20x75	10
1.4040.19.182	75x25x75	10
1.4040.19.196	75x32x75	10
1.4040.19.210	75x40x75	10
1.4040.19.212	75x50x75	10
1.4040.19.214	75x63x75	10
1.4040.19.218	90x50x90	4
1.4040.19.220	90x63x90	4
1.4040.19.224	110x50x110	3
1.4040.19.226	110x63x110	2

PP-R CROSS TE — قسام رباعي صليب حراري

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.21.020	20	250
1.4040.21.025	25	100
1.4040.21.032	32	60



PP-R TE BRANCH — قسام خماسي حراري

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.20.012	20x20	100
1.4040.20.014	25x25	60
1.4040.20.015	25x20	60



PP-R CROSS TE BRANCH — قسام زاوية حراري

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.22.020	32x20	80
1.4040.22.025	32x25	80
1.4040.22.032	40x20	60
1.4040.22.040	40x25	60



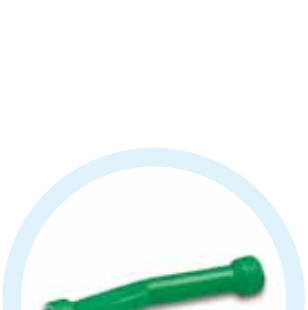
PP-R BLANKING CAP — الطبقات

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.28.020	20	900
1.4040.28.025	25	500
1.4040.28.032	32	300
1.4040.28.040	40	150
1.4040.28.050	50	80
1.4040.28.063	63	50
1.4040.28.075	75	32
1.4040.28.090	90	18
1.4040.28.110	110	9



PP-R BRIDGE WITH CONNECTIONS — جمل مع جلبة حراري

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.25.020	20	120
1.4040.25.025	25	60
1.4040.25.032	32	20



PP-R BRIDGE — جمل حراري طويل

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.26.020	20	90
1.4040.26.025	25	60
1.4040.26.032	32	40
1.4040.26.040	40	18



PP-R C BRIDGE — جمل حراري قصير

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.27.020	20	200
1.4040.27.025	25	100
1.4040.27.032	32	60



PP-R REDUCER — ناقصة حراري ذكر

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.23.010	25x20	700
1.4040.23.012	32x20	300
1.4040.23.014	32x25	350
1.4040.23.016	40x20	250
1.4040.23.018	40x25	250
1.4040.23.020	40x32	150
1.4040.23.022	50x20	150
1.4040.23.024	50x25	120
1.4040.23.026	50x32	120
1.4040.23.028	50x40	100
1.4040.23.030	63x20	80
1.4040.23.032	63x25	80
1.4040.23.034	63x32	60
1.4040.23.036	63x40	60
1.4040.23.038	63x50	50
1.4040.23.040	75x20	60
1.4040.23.042	75x25	50
1.4040.23.044	75x32	50
1.4040.23.046	75x40	50
1.4040.23.048	75x50	40
1.4040.23.050	75x63	40
1.4040.23.060	90x50	20
1.4040.23.062	90x63	20
1.4040.23.064	90x75	16
1.4040.23.066	110x63	16
1.4040.23.068	110x75	16
1.4040.23.070	110x90	12



PP-R PIPE CLIPS — كلبس ماسورة



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.29.020	20	1000
1.4040.29.025	25	800
1.4040.29.032	32	500
1.4040.29.040	40	300
1.4040.29.050	50	250

PP-R DOUBLE PIPE CLIPS — كلبس ماسورة مزدوج



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.30.020	20	400
1.4040.30.025	25	300
1.4040.30.032	32	150

PP-R END PLUG — سدة بسن



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.31.020	20	600
1.4040.31.025	25	600
1.4040.31.032	32	300
1.4040.31.U20	20 (long)	250

PP-R ELBOW FEMALE — كوع بسن انثى



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4050.16.010	20x1/2''	120
1.4050.16.012	20x3/4''	80
1.4050.16.014	25x1/2''	100
1.4050.16.016	25x3/4''	100
1.4050.16.018	32x1/2''	40
1.4050.16.020	32x3/4''	40
1.4050.16.022	32x1''	40
1.4050.16.026	40x1 1/4''	14

PP-R ELBOW MALE — كوع بسن ذكر



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4050.18.010	20x1/2''	100
1.4050.18.012	20x3/4''	80
1.4050.18.014	25x1/2''	80
1.4050.18.016	25x3/4''	80
1.4050.18.018	32x1/2''	40
1.4050.18.020	32x3/4''	40
1.4050.18.022	32x1''	40
1.4050.18.026	40x1 1/4''	16

PP-R WALLPLATE ELBOW FEMALE — كوع بسن جداري



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4050.20.010	20x1/2''	75
1.4050.20.012	25x1/2"	45
1.4050.20.014	25x3/4"	50

PP-R DOUBLE WALLPATE ELBOW FEMALE FOR WC موزع خطين أنثى

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4050.19.020	20x1/2''	20
1.4050.19.025	25x1/2"	20



PP-R DOUBLE WALLPLATE ELBOW FEMALE ميزان خلاط

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4050.21.C20	20x1/2''	35
1.4050.21.C25	25x1/2''	25



PP WALL CON. ELBOW FEMALE DOUBLE (NEW MODEL) ميزان خلاط جديد

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4050.22.C20	20x1/2''	40
1.4050.22.C25	25x1/2''	20



PP-R TE FEMALE قسام بسن انثى



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4050.30.010	20x1/2”	100
1.4050.30.012	20x3/4”	60
1.4050.30.014	25x1/2”	75
1.4050.30.016	25x3/4”	60
1.4050.30.018	32x1/2”	30
1.4050.30.020	32x3/4”	40
1.4050.30.022	32x1”	30

PP-R TE MALE قسام بسن ذكر



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4050.32.010	20x1/2”	80
1.4050.32.012	20x3/4”	60
1.4050.32.014	25x1/2”	60
1.4050.32.016	25x3/4”	60
1.4050.32.018	32x1/2”	30
1.4050.32.020	32x3/4”	30
1.4050.32.022	32x1”	30

PP-R ROUND ADAPTOR FEMALE جلبة بسن انثى



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4050.40.010	20x1/2’’	150
1.4050.40.012	20x3/4’’	120
1.4050.40.014	25x1/2’’	125
1.4050.40.016	25x3/4’’	120
1.4050.40.018	32x1/2’’	75
1.4050.40.020	32x3/4’’	75
1.4050.40.022	32x1’’	60

PP-R ROUND ADAPTOR MALE جلبة بسن ذكر



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4050.41.010	20x1/2’’	120
1.4050.41.012	20x3/4’’	100
1.4050.41.014	25x1/2’’	120
1.4050.41.016	25x3/4’’	100
1.4050.41.018	32x1/2”	75
1.4050.41.020	32x3/4’’	75
1.4050.41.022	32x1’’	75

PP-R HEXAGONAL NIPPLE FEMALE — جلبة بسن انثى



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4050.42.032	32x1''	50
1.4050.42.040	40x1 1/4''	40
1.4050.42.050	50x1 1/2''	20
1.4050.42.063	63x2''	12
1.4050.42.075	75x2 1/2''	8
1.4050.42.090	90x3''	4
1.4050.42.110	110x4''	2

PP-R HEXAGONAL NIPPLE MALE — جلبة بسن ذكر



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4050.43.032	32x1''	50
1.4050.43.040	40x1 1/4''	32
1.4050.43.050	50x1 1/2''	20
1.4050.43.063	63x2''	12
1.4050.43.075	75x2 1/2''	6
1.4050.43.090	90x3"	8
1.4050.43.110	110x4''	3

PP-R FILTER FEMALE / FEMALE — فلتر انثى / انثى



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4050.58.010	20	60
1.4050.58.012	25	30
1.4050.58.014	32	25

PP-R FILTER MALE / FEMALE — فلتر ذكر / انثى



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4050.59.010	20	60
1.4050.59.012	25	30
1.4050.59.014	32	25

ELBOW WITH LOOSE NUT — كوع بسن متحرك

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4050.17.010	20 x 1/2”	120
1.4050.17.012	20 x 3/4”	70
1.4050.17.014	25 x 3/4”	80
1.4050.17.016	25 x 1”	50
1.4050.17.017	32 x 1”	30
1.4050.17.018	32 x 1 1/4”	30



NIPPLE WITH LOOSE NUT — جلبة بسن متحرك

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4050.39.010	20 x 1/2”	180
1.4050.39.012	20 x 3/4”	150
1.4050.39.014	25 x 3/4”	120
1.4050.39.016	25 x 1”	60
1.4050.39.018	32x1”	60
1.4050.39.020	32 x 1 1/4”	40



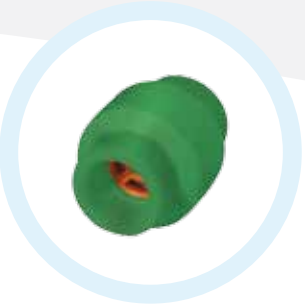
TEE WITH LOOSE NUT — قسام بسن متحرك

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4050.33.010	20 x 1/2” x 20	100
1.4050.33.012	20 x 3/4” x 20	50
1.4050.33.014	25 x 3/4” x 25	50
1.4050.33.016	25 x 1” x 25	40
1.4050.33.017	32x1”x32	20
1.4050.33.018	32 x 1 1/4” x 32	20



PP-R CHECK VALVE — صمام عدم رجوع

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4040.32.020	20	150
1.4040.32.025	25	150



PP-R TRANSITION UNION FEMALE
شد وصل انثى



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4060.13.010	20x1/2 (ECO)	200
1.4060.13.014	25x3/4 (ECO)	100
1.4060.13.020	32x1 (ECO)	75
1.4060.13.024	40x1 1/4 (ECO)	45
1.4060.11.026	50x1 1/2	25
1.4060.11.028	63x2	12
1.4060.11.030	75x2 1/2	6
1.4060.11.032	90 x 3”	3
1.4060.11.034	110x4”	2

PP-R TRANSITION UNION MALE
شد وصل ذكر



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4060.14.010	20x1/2 (ECO)	150
1.4060.14.014	25x3/4 (ECO)	100
1.4060.14.020	32x1 (ECO)	60
1.4060.14.024	40x1 1/4 (ECO)	40
1.4060.12.026	50x1 1/2	20
1.4060.12.028	63x2	12
1.4060.12.030	75x2 1/2	6
1.4060.12.032	90x3”	3
1.4060.12.034	110x4”	2

PP-R BALL VALVE WITH DOUBLE TRANSITION UNION

محبس سكين شد وصل من الطرفين



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4070.10.020	20	30
1.4070.10.025	25	25
1.4070.10.032	32	15

PP-R BALL VALVE WITH SINGLE TRANSITION UNION

محبس سكين شد وصل من طرف واحد



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4070.11.020	20	35
1.4070.11.025	25	30
1.4070.11.032	32	15

PP-R CHROME VALVE LUXURY

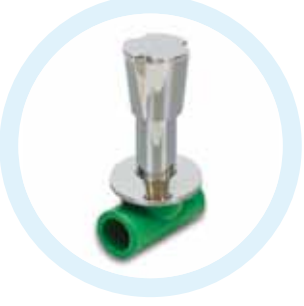
محبس كروم كبشة مثلثة



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4070.13.020	20	20
1.4070.13.025	25	16
1.4070.13.032	32	15

PP-R CHROME VALVE WITH TREFOIL

محبي كروم كبشة شجرة



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4070.32.020	20	20
1.4070.32.025	25	16
1.4070.32.032	32	15

PP-R CHROME VALVE HIDDEN — محبس كروم مخفي قصير

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4070.14.020	20	35
1.4070.14.025	25	25
1.4070.14.032	32	20



PP-R BALL VALVE — محبس سكين مدور

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4070.15.020	20	40
1.4070.15.025	25	35
1.4070.15.032	32	20
1.4070.15.040	40	15
1.4070.15.050	50	8
1.4070.15.063	63	5
1.4070.15.075	75	5



PP-R BALL VALVE FOR RADIATOR · — محبس سكين كوع

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4070.18.020	20x1/2''	50
1.4070.18.025	25x3/4''	30



PP-R BALL VALVE FOR RADIATOR — محبس سكين ذكر

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4070.19.020	20x1/2''	50
1.4070.19.025	25x3/4''	40



PP-R SHUT OFF VALVE - Metal Handle
محبس سكين - قبضة المعدني

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4070.21.020	20	30
1.4070.21.025	25	24
1.4070.21.032	32	15
1.4070.21.040	40	10



PP-R SHUT OFF VALVE - Plastic Handle
محبس سكين - قبضة بلاستيكي

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4070.20.020	20	30
1.4070.20.025	25	24
1.4070.20.032	32	15
1.4070.20.040	40	10



PP-R BALL VALVE (LONG)
محبس سكين طويل

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4070.24.020	20	35
1.4070.24.025	25	25



PP-R BALL VALVE FOR RADIATOR WITH CORE ELBOW
محبس سكين مدور كوع

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4070.26.020	20x1/2	40

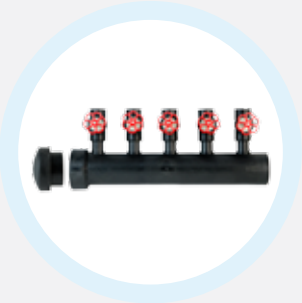


PP-R BALL VALVE FOR RADIATOR WITH CORE
محبس سكين مدور ذكر

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
1.4070.27.020	20	40



PP90 MANIFOLD — شبكة جاهزة ٩٠ مم



الرمز Code	المقاس ملم d (mm)	التعبئة
3B65-MNF-900332	PP 90 Manifold 3 ways Right Black - Berke	
3B65-MNF-900432	PP 90 Manifold 4 ways Right Black - Berke	
3B65-MNF-900532	PP 90 Manifold 5 ways Right Black - Berke	
3B65-MNF-901332	PP 90 Manifold 3 ways Left Black - Berke	
3B65-MNF-901432	PP 90 Manifold 4 ways Left Black - Berke	
3B65-MNF-901532	PP 90 Manifold 5 ways Left Black - Berke	

WELDING SET — مكينة لحام حراري



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
4400.10.015	20-40	5

WELDING MACHINE — ماكينة اللحام



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
4400.10.022	90-110	1

WELDING TOOLS — لقمة لحام



الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
4400.11.010	20	100
4400.11.012	25	100
4400.11.014	32	50
4400.11.016	40	40
4400.11.018	50	30
4400.11.020	63	20
4400.11.022	75	12
4400.11.024	90	5
4400.11.026	110	5

FOIL PEELER
مقشط ماسورة

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
4400.15.010	20-25	35
4400.15.012	32-40	20
4400.15.014	50-63	10
4400.15.016	75	10
4400.15.020	90	4
4400.15.022	110	4



PIPE SCISSORS
مقص ماسورة

الرمز Code	المقاس ملم d (mm)	التعبئة Package Standard
4400.14.010	20-40	20
4400.14.012	32-63	5





لا تعرض المواسير والاكسسوارات لأشعة الشمس . ضع المواسير في اماكن بعيدة عن المواد الحادة والخشنة . لا تستعمل المواسير المتضررة .



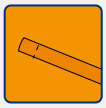
لا تستخدم النار لتسخين المواسير . استخدم الهواء الساخن لثني المواسير .



يجب تنظيف المواسير من الغبار قبل تركيبها .



قص المواسير بالمقص الخاص ولا تقطعها بالمواد الحادة الاخرى لانها تسبب الشوائب



حدد طول مكان اللحام قبل البدء باللحام



عند القيام باللحام اتبع التعليمات الموجودة في كتلوج المصنع (الحرارة , فترة التسخين , وغيره)



لا تقم بلف الماسورة او الاكسسوارات اثناء عملية اللحام



لا تستخدم مواسير البولي بروبلين واكسسواراتها للمياه القابلة للتجمد فان تمدد المياه قد يتسبب بكسر الماسورة



ضع منتجات البولي بروبلين في مكان آمن في حال انخفاض درجة الحرارة مما يضعف مقاومة مواد البروبلين وتصبح هشة ولا تتحمل الصدمات



اقصى درجة يمكن لف المواسير والاكسسوار بعد اللحام هي خمس درجات

بعد قص المواسير متعددة الطبقات يجب قشط الزوائد الحادة قبل التركيب .

بعد قص المواسير التي تحتوي على طبقة من الامنيوم يجب الحرص على ازالة الشوائب العالقة من الالمنيوم والا قد تتسبب بتسريب المياه

لتجنب تسرب المياه اثناء عملية التركيب استخدم لاصق التفلون مع الاكسسوارات المترابطة

قبل تغطية التمديدات بشكل كامل قم بضغط الشبكة باستخدام مياه درجة حرارتها مساوية لدرجة حرارة الغرفة و تأكد من عدم وجود اي هواء داخل الشبكة . ابقي الشبكة مضغوطة لفترة ٢٤ ساعة مستخدما ضغط لا يتجاوز ٢٠ بار تحت المراقبة وبعد انتهاء فترة ٢٤ ساعة قم بفحص الوصلات للتأكد من عدم وجود اي تسريب منها .



СИСТЕМА СЕРТИФИКАЦИИ ГОСТ Р
ГОССТАНДАРТ РОССИИ
СЕРТИФИКАТ СООТВЕТСТВИЯ
№ РОСС TR.AE80.H00346
Срок действия С 08.02.2008 по 07.02.2011
0798503
ОРГАН ПО СЕРТИФИКАЦИИ рег. № РОСС RU.0001.11АЕ80
ПРОДУКЦИЯ Трубы и соединительные части к ним из полипропилена,
СООТВЕТСТВУЕТ ТРЕБОВАНИЯМ НОРМАТИВНЫХ ДОКУМЕНТОВ
ГОСТ Р 52134-2003
ИЗГОТОВИТЕЛЬ "BERKE PLASTIK SANAYI VE TICARET A. S."
СЕРТИФИКАТ ВЫДАН "BERKE PLASTIK SANAYI VE TICARET A. S."
НА ОСНОВАНИИ протоколов испытаний
ФЕДЕРАЛЬНАЯ СЛУЖБА ПО НАДЗОРУ В СФЕРЕ ЗАЩИТЫ ПРАВ ПОТРЕБИТЕЛЕЙ И БЛАГОПОЛУЧИЯ ЧЕЛОВЕКА
САНИТАРНО-ЭПИДЕМИОЛОГИЧЕСКОЕ ЗАКЛЮЧЕНИЕ
№ 77.01.16.229.П.007454.02.08 от 06.02.2008
ZERTIFIKAT
SKZ
Certificate
The SKZ - TeCoKA GmbH awards the following company
Berke Plastik Sanayi Ticaret A.S.
Semsettin Gunaltay Cad. No. 16/1
Bostanci
34744 Kadikoy / Istanbul
TURKEY
the right to use the SKZ - testing and inspection mark
A 410
for the following plastic products
Heating pipes made of polypropylene PP-R 80
DVGW-Baumusterprüfzertifikat
DVGW type examination certificate
DVGW
№ 247078
СЕРТИФІКАТ ВІДПОВІДНОСТІ
UA1.081.0193780-08
30 грудня 2008 до 29 грудня 2013
ROMÂNIA
MINISTERUL DEZVOLTARII REGIONALE SI TURISMULUI
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AVIZ TEHNIC
ТЕХНИЧЕСКОЕ СВИДЕТЕЛЬСТВО
ТС 01.0950.10
KALİTE YÖNETİM SİSTEMİ BELGESİ
QUALITY MANAGEMENT SYSTEM CERTIFICATE
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PPR
Technical Catalogue

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